

Overlapping Bullet Hole Detection Based on Improved Watershed Algorithm

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Abstract—The automatic target reporting system based on image processing has a high rate of missed detection of overlapping bullet holes on the target plate, which will identify the overlapping bullet holes as one bullet hole, for which an overlapping bullet hole detection method based on improved watershed algorithm is proposed. The improved watershed algorithm performs area-based detection of the connected area and background of the bullet hole to segment the bullet hole and background and eliminate the isolated area. For the segmented overlapping bullet holes, a chain code tracking method is used to obtain their boundaries, and finally a least-squares fitting of the boundary information is performed to obtain the bullet hole center of mass in the overlapping hole area. The results show that compared with the traditional watershed algorithm, the proposed improved watershed algorithm can effectively segment the overlapping bullet holes, and for the localization of the bullet hole center of mass, the measurement error for the heavy holes is no more than 3 pixels when the overlap degree does not exceed 90.43%.

Keywords—overlapping bullet holes, improved watershed algorithm, chain code tracking, least squares

I. INTRODUCTION

An overlapping bullet hole is an area with a certain overlapping area at the location where multiple rounds land on the target paper when fired. In the target reporting system, the recognition ability for overlapping holes is a factor to measure the reliability and accuracy of a system [1]. With the development of image processing technology, some mature recognition methods have emerged [2]. Ni, Jinping et al. proposed a heavy hole recognition method based on subtraction technique, which subtracts the acquired front and rear target surface images to obtain the bullet hole information and identifies the location of the heavy hole by finding the location of the center of mass of the bullet hole through the regional

center of gravity method [3], but the method is influenced by the environment and poorly handles the edges of the bullet hole in complex backgrounds. Roger et al. proposed a heavy hole identification method based on the existence probability map [4], which maps the intersection points of the intersecting regions of heavy holes in the processed images to the peak region of the circular probability map, and identifies the bullet hole location by screening the peak to determine the parameters of the bullet hole. Since its recognition quality depends on the selection of edge points, while more edge points will lead to an increase in computation time. Song Yu et al. proposed an improved maximum interclass variance based method [5] to find the best threshold value by segmenting the bullet hole gradient image thus reducing the false segmentation caused by the traditional Otsu method, while using the minimum outer rectangle to locate the center-of-mass coordinates of the heavy holes. However, this method still produces false segmentation for the ring value, ring line and other regions. The above methods do not effectively solve the heavy hole problem.

In this paper, we propose an overlapping hole identification method based on improved watershed algorithm. The processed image is threshold segmented to reduce its background noise, the overlapping bullet hole area is extracted by the improved watershed algorithm, and finally the center-of-mass position of multiple bullet holes in the bullet hole area is obtained using the boundary tracking method. This method can have better segmentation effect and higher accuracy for the part of irregular bullet hole edge and connected between bullet holes.

II. IMPROVED OVERLAPPING BULLET HOLES FOR WATERSHED COUNTING

A. Traditional Watershed Algorithm

The traditional watershed algorithm is a mathematical morphological segmentation method based on topological

theory. The idea is to represent the grayscale values of pixels in an image as geodetic elevation, and there are local areas of the image with minimal values, which are called catchment basins, and the boundary between the catchment basins is the watershed. The concept of a watershed can be understood as the injection of water into each catchment basin, which flows to the basin where the regional minima are located, and as the water level rises, it is necessary to raise the "dikes" to avoid connecting the various catchment basins, and these "dikes" are the dividing lines of the image [6] [7].

In this paper, we improve on the watershed algorithm based on the distance transform. The distance transform describes the magnitude of the distance between a pixel point and a region in an image, and its can be used to represent a binary image in a gray scale map. The basic flow of the watershed algorithm based on image transform is as follows [8][9]:

- 1) *Image pre-processing*: the acquired image is pre-processed to improve its local contrast and make the image details more obvious;
- 2) *Open and closed operations*: morphological open and closed operations on pre-processed images to eliminate small protrusions, fill holes and obtain foreground and background markers;
- 3) *Marking foreground objects*: for reconstructed objects create extreme values in their internal regions;
- 4) *Calculate background marker*: Binary operation is performed on the image, and the watershed transformation is performed to obtain the dividing line of adjacent areas to form the background marker;
- 5) *Image segmentation*: Image segmentation is performed by modifying the gradient image to have the minimum value of the region appear on the obtained marker;
- 6) *Output segmented image*: superimpose foreground and background markers onto the image for output.

B. Improved Watershed Algorithm

The traditional watershed algorithm produces noise in the target image due to the target paper and the environment; and also segments the ring line, the ring value and the area of the bullet hole to be extracted, which will have an impact on the subsequent round center estimation process. In order to improve the segmentation effect, this paper makes the following improvements on the distance-transformed watershed algorithm: after the pre-processing of grayscale transformation, filtering and binarization of the acquired image, the area-based target detection is performed on the connected areas of the image to eliminate the noisy areas and the ring line and ring value areas.

By the algorithm based on the area of the region, the bullet hole area of the acquired image can be detected and extracted, thus improving the effectiveness of the watershed algorithm segmentation [10]. Furthermore, based on the area of the connected region

to determine whether the bullet hole is a heavy hole or a single hole, the bullet hole will be displayed on the target paper as a class circle with a diameter of 6 pixels. The area of the connected region can be computed to determine the hole case.

The implementation process is shown in Fig. 1 with the following steps:

- 1) Pre-processing of the image of the acquired target paper;
- 2) To reduce the mis-segmentation of the watershed algorithm, the image is binarized by setting a gray scale threshold;
- 3) Morphological opening and closing operations on binary images in order to eliminate the effects of fine, tiny protrusions and holes;
- 4) Select the region of interest in the image; calculate the area of the connected components within the region and compare it with the set threshold (area of a single bullet hole pixel) to eliminate the noisy area generated by the target paper itself; and perform watershed segmentation if it is greater than the set threshold, and set the values of the pixels within the region to zero if it is less than the threshold;;
- 5) Watershed splitting of heavy holes in the area to obtain overlapping bullet hole areas.

According to the above algorithm can be achieved to obtain the heavy hole image after watershed segmentation to get the bullet hole area, and at the same time based on the area area detection of isolated areas to eliminate.

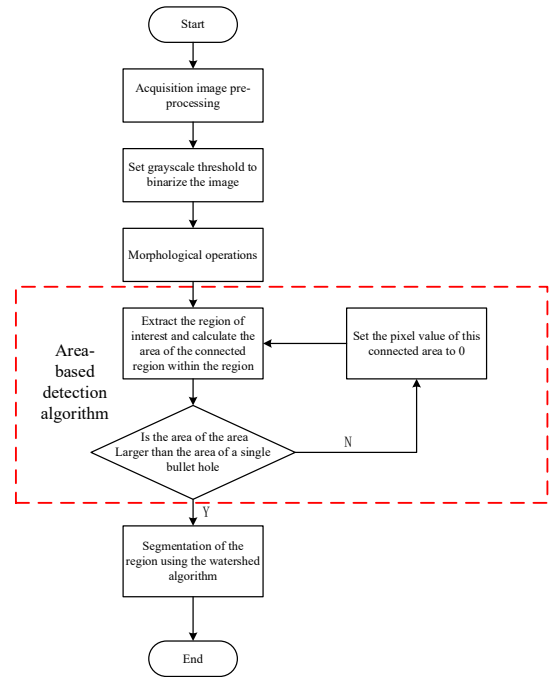


Fig. 1. Flowchart of the improved watershed algorithm

III. ESTIMATION OF HEAVY HOLE CIRCLE CENTER

Overlapping bullet holes are formed by the partial overlap of multiple class circles. Hence, in order to be able to detect the number of overlapping holes, the circle center of the overlapping holes needs to be estimated. The algorithm used is the boundary chain code tracking algorithm of eight neighbors. In other words, the chain code of its boundary is obtained by tracking and searching along the direction of eight pixels adjacent to the pixel point; then the circle center is determined by the least squares;

and then the bullet hole estimate is obtained by the circle center and radius.

A. Boundary Extraction

For image boundary extraction, the pixel points adjacent to the current pixel are called the eight neighboring pixels of that pixel, as shown in Fig. 2 [11]. It is assumed that the target pixel C is to the right of the adjacent point labeled as 0; and the adjacent points are numbered separately in counterclockwise direction.

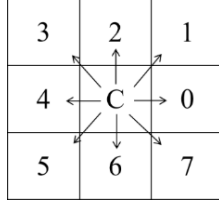


Fig. 2. Eight Neighborhoods

The eight-neighborhood boundary tracking algorithm completes the extraction of the boundary in three main steps as follows:

1) Determine the starting point of the search boundary; in this paper, the boundary points are scanned from left to right and from top to bottom, and the first boundary point is determined as the starting point of the target boundary;

2) Boundary discrimination criterion; the chain code makes a judgment on whether a pixel in the neighborhood is a boundary point according to the clockwise direction, and when a point is confirmed to be the target boundary point, the boundary point is recorded and the pixel point where it is located is updated to repeat the tracking process;

3) The termination criterion; for the case of the boundary extraction of the bullet hole, the boundary extraction is considered complete when the tracked pixel point is the starting point and there is no different chain code, as shown in Fig. 3.

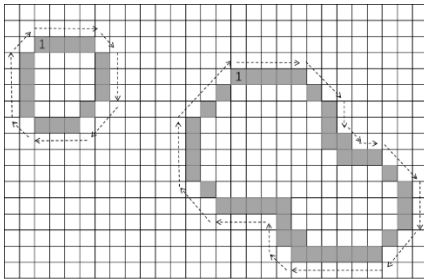


Fig. 3. Guidelines for termination of bullet holes

B. Least Squares Fitting

The boundary of the circle-like bullet hole can be obtained after tracing through the chain code, at this point the boundary is the case of containing both single and heavy holes. The boundary data are next fitted using least squares to obtain the estimated circle center and radius of the circle [12].

According to the boundary tracking algorithm can obtain a series of boundary points data $\{x_i, y_i\}$, assuming that the

estimated circle center is (x_c, y_c) . According to the equation of the circle we can get the following equation:

$$(x - x_c)^2 + (y - y_c)^2 = R^2 \quad (1)$$

For the fitted data, the estimated circle center is fixed and can be made to be:

$$\begin{cases} u = -2x_c \\ v = -2y_c \\ s = x_c^2 + y_c^2 - R^2 \end{cases} \quad (2)$$

From (1) and (2), we have:

$$x^2 + y^2 + ux + vy + s = 0 \quad (3)$$

Based on the boundary point data $\{x_i, y_i\}$ (3) can be written in matrix form as follows:

$$\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ \vdots & \vdots & \vdots \\ x_n & y_n & 1 \end{bmatrix} \cdot \begin{bmatrix} u \\ v \\ s \end{bmatrix} = \begin{bmatrix} -x_1^2 - y_1^2 \\ -x_2^2 - y_2^2 \\ \vdots \\ -x_n^2 - y_n^2 \end{bmatrix} \quad (4)$$

From (4) can be written as:

$$X \cdot A = Y \quad (5)$$

The matrix A can be obtained according to the least squares method as:

$$A = (X^T X)^{-1} X^T Y \quad (6)$$

The coordinates and radius of the estimated circle center can be obtained from (2) and (6):

$$\begin{cases} x_c = -\frac{u}{2} \\ y_c = -\frac{v}{2} \\ R = \sqrt{x_c^2 + y_c^2 - s} \end{cases} \quad (7)$$

The radius between the edge of the bullet hole and the estimated center of the circle can be obtained from the distance from the coordinates of the boundary point to the estimated center of the circle based on the estimated center of the overlapping bullet hole as the center of mass point.

$$R'_i = \sqrt{(x_i - x_c)^2 + (y_i - y_c)^2} \quad (8)$$

where R'_i denotes the distance between each point of the edge and the center of the fitted circle.

$$\Delta r_i = R - R'_i \quad (i=1, 2, \dots, n) \quad (9)$$

Since the radius of each point under the same fitting circle fluctuates within a certain range from the fitting radius, it is possible to distinguish whether the boundary point data are

under the same fitting circle by setting a threshold value, so as to determine whether the obtained boundary is a case of overlapping bullet holes or single holes.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Analysis of Segmentation Results for Heavy Holes

For the acquired reperforated images, the median filtering process will be carried out first to filter out the noise in the images initially, and then the traditional watershed algorithm and the improved watershed algorithm will be used to process them respectively, and the processing results are shown in Fig. 4.

In this paper, two evaluation metrics, Intersection over Union (IOU) and Accuracy (Acc), are used to evaluate the segmented bullet hole regions. IOU indicates the ratio of intersection and concurrence between segmented and true results, and higher IOU indicates better segmentation. Acc is the ratio of correct segmentation to total pixels in the segmented results.

The evaluation of the segmentation results of the traditional watershed algorithm and the algorithm with improved watershed are shown in Table I.

TABLE I. COMPARISON OF SEGMENTATION RESULTS BETWEEN THE HEAVY-HOLE TRADITIONAL WATERSHED ALGORITHM AND THE IMPROVED WATERSHED ALGORITHM

Serial number	Traditional Watershed Algorithm		Improved Watershed Algorithm	
	IOU	ACC	IOU	ACC
1	0.856	0.894	0.931	0.915
2	0.893	0.902	0.945	0.947
3	0.904	0.921	0.962	0.972

Fig. 4(a), (b), and (c) are the images of the bullet holes segmented by the traditional watershed algorithm, and there are some isolated areas on the images, which are caused by the background, while Fig. 4(d), (e), and (f) are the bullet holes segmented by the improved watershed algorithm, and it can be seen that the segmentation effect is better and the isolated areas are eliminated. Based on the comparison of the two methods, it is easy to see that the improved watershed algorithm can better

clarify the isolated regions outside the bullet holes compared to the traditional method.

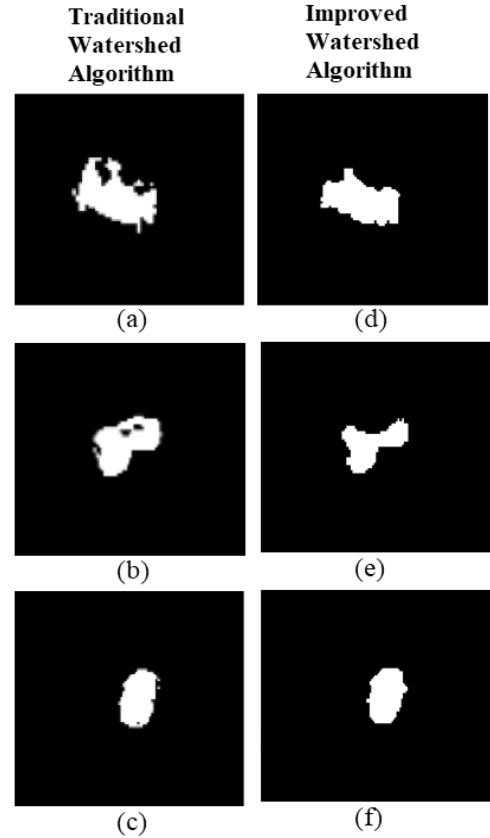


Fig. 4. Comparison of segmentation effect of overlapping bullet holes

B. Analysis of the Effect of Heavy Hole Circle Center and Coordinate Positioning

In order to check the accuracy of the least-squares circle method for detecting the center of mass of heavy holes, this paper compares the coordinates of the center of the circle obtained by the least-squares circle method with the coordinates of the center of mass of the actual bullet hole for the 10 sets of images of heavy holes obtained, so as to obtain the measurement error of the method. The results are shown in Table II.

TABLE II. BULLET HOLE CENTER OF MASS POSITION MEASUREMENT ERROR

Serial number	Bullet hole 1		Measurement error	Bullet hole 2		Measurement error	Overlap
	Actual Coordinates	Measurement coordinates		Actual Coordinates	Measurement coordinates		
1	(170,373)	(171,372)	1.41	(175,375)	(173,375)	2.00	38.84%
2	(213,400)	(212,398)	2.24	(217,400)	(216,401)	1.41	68.01%
3	(308,381)	(308,381)	0	(305,390)	(305,390)	0	6.41%
4	(339,349)	(340,349)	1.00	(345,344)	(345,345)	1.00	18.93%
5	(361,295)	(362,297)	2.24	(364,298)	(366,300)	2.83	59.82%
6	(358,218)	(360,218)	2.00	(357,213)	(358,212)	1.41	41.59%
7	(335,180)	(337,182)	2.83	(333,177)	(330,179)	3.61	74.51%
8	(197,156)	(197,155)	1.00	(193,161)	(192,160)	1.41	27.25%
9	(202,213)	(200,211)	2.84	(201,210)	(203,212)	2.84	85.78%
10	(186,243)	(185,245)	2.23	(183,247)	(182,245)	2.24	42.14%

Due to the difficulty of obtaining reperforation data, in order to ensure that sufficient reperforation data are available for the experiments, this paper uses simulation to set up 200 sets of reperforation experiments to verify the accuracy of the method in this paper.

The software used in this paper is python 3.9, opencv = 4.7.0, with a 64-bit operating system and a computer configuration of Inter Core-i7-7700, 2.8Ghz, 16GB RAM.

The program was set up with 200 sets of reperforation images of two bullet holes with overlap from 5% to 95%, and the measurement error of the center of mass of the bullet hole was obtained by calculating its center and radius through the least squares circle as shown in Fig. 5.

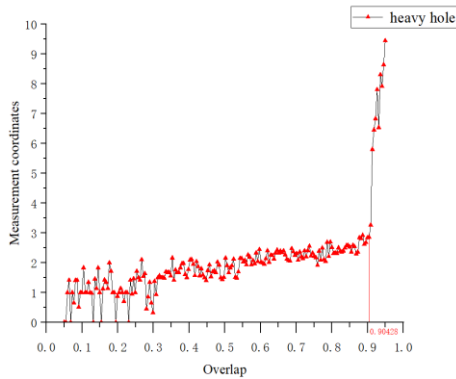


Fig. 5. Bullet hole center of mass measurement error

From the data results, the measurement error of overlapping bullet holes obtained by using the least squares circle method is at a small level when the degree of overlap is lower than 86.36%, and the measurement error gradually increases when the degree of overlap is higher than 86.36%, especially when it is greater than 90.43%, and the accuracy of the least squares circle method for identifying overlapping bullet holes decreases significantly.

V. CONCLUSION

In this paper, the identification of overlapping bullet holes is investigated based on the improved watershed algorithm. The accuracy of the method is verified by analyzing the acquired overlapping bullet hole images. Firstly, the acquired images are pre-processed, secondly, the area-based watershed algorithm is used to separate the bullet holes from the background to obtain the overlapping images of the bullet holes and remove the influence of loop lines and loop values, then the segmented bullet hole images are boundary traced to obtain the contour data, and finally, the least squares process is performed on the circle centers and radii of the bullet holes based on the contour information, since there are two or more circle centers for

overlapping bullet holes. Therefore, a judgment on the circle centers can determine whether they are overlapping holes. The method of this paper is verified by 200 sets of overlapping bullet holes with different degrees of overlap, and the following conclusions are obtained:

- For the image of overlapping bullet holes based on the improved watershed algorithm can effectively segment the background and overlapping bullet holes partly, compared with the traditional watershed algorithm in the single hole segmentation effect has been greatly improved.
- The least squares method was used to fit the bullet hole edge data to obtain the estimated circle center and radius, and it was determined experimentally that the measurement error would not exceed 3 pixels for heavy holes with no more than 90.43% overlap, resulting in more accurate measurements.

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